

INDEFINITE INTEGRATION

- $\int K dx = Kx + C$, where $K \in R$.
- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ provided $n \neq -1$
- $\int |x|^n dx = \frac{|x|^{n+1}}{n+1} + C$ provided $n \neq -1$
- $\int \frac{1}{x} dx = \log|x| + C$.
- $\int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C$
- $\int e^x dx = e^x + C$
- $\int a^x dx = \frac{a^x}{\log a} + C$ for $a > 0, a \neq 1$
- $\int \sqrt{x} dx = \frac{2}{3} x\sqrt{x} + C$
- $\int \sin x dx = -\cos x + C$
- $\int \cos x dx = \sin x + C$
- $\int \tan x dx = -\log|\cos x| + C = \log|\sec x| + C$
- $\int \cot x dx = -\log|\cos ecx| + C = \log|\sin x| + C$
- $\int \sec x dx = \log|\sec x + \tan x| + C = \log\left|\tan\left(\frac{p}{4} + \frac{x}{2}\right)\right|$
- $\int \cos ecx dx = \log|\cos ecx - \cot x| + C = -\log|\cos ecx + \cot x| + C = \log\left|\tan \frac{x}{2}\right| + C$
- $\int \sec x \tan x dx = \sec x + C$
- $\int \cos ecx \cot x dx = -\cos ecx + C$
- $\int \sec^2 x dx = \tan x + C$
- $\int \cos ec^2 x dx = -\cot x + C$
- $\int \sinh x dx = \cosh x + C$
- $\int \cosh x dx = \sinh x + C$
- $\int \tanh x dx = \log|\cosh x| + C$
- $\int \coth x dx = \log|\sinh x| + C$
- $\int \sec h x dx = 2 \tan^{-1}(e^x) + C$

- $\int \operatorname{sech} x dx = \log \left| \tanh \frac{x}{2} \right| + C$
- $\int \sec hx \tanh x dx = -\sec hx + C$
- $\int \operatorname{sech} x \coth x dx = -\operatorname{sech} x + C$
- $\int \sec h^2 x dx = \tanh x + C$
- $\int \operatorname{sech}^2 x dx = -\coth x + C$
- $\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C = -\cos^{-1} x + C$
- $\int \frac{1}{1+x^2} dx = \tan^{-1} x + C = -\cot^{-1} x + C$
- $\int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1} x + C = -\operatorname{cosec}^{-1} x + C$
- $\int \frac{1}{\sqrt{1+x^2}} dx = \sinh^{-1} x + C$
- $\int \frac{1}{\sqrt{x^2-1}} dx = \cosh^{-1} x + C$
- $\int f'(ax+b) dx = \frac{f(ax+b)}{a} + C$
- $\int \frac{f'(x)}{f(x)} dx = \log |f(x)| + C$
- $\int \frac{f'(x)}{\sqrt{f(x)}} dx = 2\sqrt{f(x)} + C$
- If $\int f(x) dx = F(x) + C$, then $\int f(g(x))g'(x) dx = F(g(x)) + C$
- $\int (f(x))^n f'(x) dx = \frac{(f(x))^{n+1}}{n+1} + C$
- $\int f'(g(x))g'(x) dx = f(g(x)) + C$
- $\frac{d}{dx} \int f(x) dx = f(x)$
- $\int \frac{d}{dx} (f(x)) dx = f(x) + C$
- $\int \frac{1}{\sqrt{a^2-x^2}} dx = \sin^{-1} \frac{x}{a} + C$
- $\int \frac{1}{\sqrt{x^2-a^2}} dx = \cosh^{-1} \frac{x}{a} + C = \log \left| x + \sqrt{x^2-a^2} \right| + C$
- $\int \frac{1}{\sqrt{x^2+a^2}} dx = \sinh^{-1} \frac{x}{a} + C = \log \left| x + \sqrt{x^2+a^2} \right| + C$

- $\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$
- $\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$
- $\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \log \left| \frac{a-x}{a+x} \right| + C$
- $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$
- $\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \cosh^{-1} \frac{x}{a} + C = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log \left| x + \sqrt{x^2 - a^2} \right| + C$
- $\int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \sinh^{-1} \frac{x}{a} + C = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log \left| x + \sqrt{x^2 + a^2} \right| + C$
- $\int \frac{f'(x)}{\sqrt{a^2 - (f(x))^2}} dx = \sin^{-1} \frac{f(x)}{a} + C$
- $\int \frac{f'(x)}{\sqrt{(f(x))^2 - a^2}} dx = \cosh^{-1} \frac{f(x)}{a} + C = \log \left| f(x) + \sqrt{(f(x))^2 - a^2} \right| + C$
- $\int \frac{f'(x)}{\sqrt{(f(x))^2 + a^2}} dx = \sinh^{-1} \frac{f(x)}{a} + C = \log \left| f(x) + \sqrt{(f(x))^2 + a^2} \right| + C$
- $\int \frac{1}{(x+a)(x+b)} dx = \frac{1}{b-a} \log \left| \frac{x+a}{x+b} \right| + C$
- $\int \frac{1}{(ax+b)(cx+d)} dx = \frac{1}{ad-bc} \log \left| \frac{ax+b}{cx+d} \right| + C$
- $\int \frac{1}{(x^2 + a^2)(x^2 + b^2)} dx = \frac{1}{b^2 - a^2} \left[\frac{1}{a} \tan^{-1} \frac{x}{a} - \frac{1}{b} \tan^{-1} \frac{x}{b} \right] + C$
- $\int \frac{x}{(x^2 + a^2)(x^2 + b^2)} dx = \frac{1}{2(b^2 - a^2)} \log \left| \frac{x^2 + a^2}{x^2 + b^2} \right| + C$
- $\int \frac{1}{x(x^n + 1)} dx = \frac{1}{n} \log \left| \frac{x^n}{x^n + 1} \right| + C$
- $\int \frac{1}{x(x^n - 1)} dx = \frac{1}{n} \log \left| \frac{x^n - 1}{x^n} \right| + C$
- $\int \frac{1}{x(1 - x^n)} dx = \frac{1}{n} \log \left| \frac{x^n}{1 - x^n} \right| + C$

- $\int f(x)g(x)dx = f(x)\int g(x)dx - \int \left(\left(\int g(x)dx \right) \times \frac{d}{dx}(f(x)) \right) dx$ where $f(x)$ is the first function and $g(x)$ is the second function.
 - First function can be chosen by the following ILATE rule as preference.
 - I – Inverse Trigonometric functions
 - L – Logarithmic functions
 - A – Algebraic functions
 - T – Trigonometric functions
 - E – Exponential functions
 - If one of the two functions is not directly integrable then take this function as first function.
 - If one of the function is not directly integrable and there is no second function then take unity as the second function.
- $\int (u.v)dx = uv_1 + u^1v_2 + u^2v_3 + u^3v_4 + \dots$ where $u^1, u^2, u^3, \dots$ are 1st, 2nd, 3rd ... derivative of u and $v_1, v_2, v_3, \dots$ are 1st, 2nd, 3rd ... integration of v .
- $\int e^x (f(x) + f'(x))dx = e^x f(x) + C$
- $\int e^{-x} (f(x) - f'(x))dx = -e^{-x} f(x) + C$
- $\int e^{ax} (a.f(x) + f'(x))dx = e^{ax} f(x) + C$
- $\int e^{ax} \left(f(x) + \frac{f'(x)}{a} \right) dx = \frac{e^{ax} f(x)}{a} + C$
- $\int (xf(x) + f'(x))dx = xf(x) + C$
- $\int x^n \log x dx = \frac{x^{n+1}}{n+1} \left(\log x - \frac{1}{n+1} \right) + C$ for $n \neq -1$
- $\int e^{ax} \sin(bx+c)dx = \frac{e^{ax}}{a^2+b^2} (a \sin(bx+c) - b \cos(bx+c)) + C = \frac{e^{ax}}{a^2+b^2} \sin \left(bx+c - \tan^{-1} \frac{b}{a} \right) + C$
- $\int e^{ax} \cos(bx+c)dx = \frac{e^{ax}}{a^2+b^2} (a \cos(bx+c) + b \sin(bx+c)) + C = \frac{e^{ax}}{a^2+b^2} \cos \left(bx+c - \tan^{-1} \frac{b}{a} \right) + C$
- $\int a^x \sin(bx+c)dx = \frac{a^x}{(\log a)^2 + b^2} (\log a \sin(bx+c) - b \cos(bx+c)) + C$
- $\int a^x \cos(bx+c)dx = \frac{a^x}{(\log a)^2 + b^2} (\log a \cos(bx+c) + b \sin(bx+c)) + C$
- $\int \sin^{-1} x dx = x \sin^{-1} x + \sqrt{1-x^2} + C$
- $\int \cos^{-1} x dx = x \cos^{-1} x - \sqrt{1-x^2} + C$
- $\int \tan^{-1} x dx = x \tan^{-1} x - \frac{1}{2} \log(1+x^2) + C$
- $\int \cot^{-1} x dx = x \cot^{-1} x + \frac{1}{2} \log(1+x^2) + C$

- $\int \sec^{-1} x dx = x \sec^{-1} x - \cosh^{-1} x + C$
- $\int \cos ec^{-1} x dx = x \cos ec^{-1} x + \cosh^{-1} x + C$
- $\int \sinh^{-1} x dx = x \sinh^{-1} x - \sqrt{x^2 + 1} + C$
- $\int \cosh^{-1} x dx = x \cosh^{-1} x - \sqrt{x^2 - 1} + C$
- $\int \sinh^{-1} \frac{x}{a} dx = x \sinh^{-1} \frac{x}{a} - \sqrt{x^2 + a^2} + C$
- $\int \cosh^{-1} \frac{x}{a} dx = x \cosh^{-1} \frac{x}{a} - \sqrt{x^2 - a^2} + C$
- For $n \in N$, $I_n = \int \sin^n x dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$
- For $n \in N$, $I_n = \int \cos^n x dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} I_{n-2}$
- For $n \in N$, $I_n = \int \tan^n x dx = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$
- For $n \in N$, $I_n = \int \cot^n x dx = -\frac{\cot^{n-1} x}{n-1} - I_{n-2}$
- For $n \in N$, $I_n = \int \sec^n x dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}$
- For $n \in N$, $I_n = \int \cos ec^n x dx = -\frac{\cos ec^{n-2} x \cot x}{n-1} + \frac{n-2}{n-1} I_{n-2}$
- For $n \in N$, $I_n = \int x^n e^{ax} dx = \frac{e^{ax} x^n}{a} - \frac{n}{a} I_{n-1}$
- For $n \in N$, $I_n = \int (\log x)^n dx = x(\log x)^n - n I_{n-1}$
- For $m, n \in N$, $I_{m,n} = \int \sin^m x \cos^n x dx = -\frac{\sin^{m+1} x \cos^{n-1} x}{m+n} + \frac{m-1}{m+n} I_{m-2,n}$
- For $m, n \in N$, $I_{m,n} = \int x^m (\log x)^n dx = \frac{x^{m+1} (\log x)^n}{m+1} - \frac{n}{m+1} I_{m,n-1}$
- **Some Standard Substitutions:**

- (1) For terms of the form $x^2 + a^2$ or $\sqrt{x^2 + a^2}$ put $x = a \tan q$ or $x = a \cot q$.
- (2) For terms of the form $x^2 - a^2$ or $\sqrt{x^2 - a^2}$ put $x = a \sec q$ or $x = a \cos ec q$.
- (3) For terms of the form $a^2 - x^2$ or $\sqrt{a^2 - x^2}$ put $x = a \sin q$ or $x = a \cos q$.
- (4) If both $\sqrt{a-x}$ and $\sqrt{a+x}$ are present then put $x = a \cos q$.
- (5) For the type $\sqrt{(x-a)(b-x)}$ or $\sqrt{\frac{(x-a)}{(b-x)}}$ put $x = a \cos^2 q + b \sin^2 q$.

- (6) For the type $\sqrt{(x-a)(x-b)}$ or $\sqrt{\frac{(x-a)}{(x-b)}}$ put $x = a \sec^2 q - b \tan^2 q$.
- (7) For the type $(\sqrt{x^2 + a^2} \pm x)^n$ or $(x \pm \sqrt{x^2 + a^2})^n$ or $\frac{1}{(x \pm \sqrt{x^2 + a^2})^n}$ or $\frac{(x \pm \sqrt{x^2 + a^2})^n}{x^2 + a^2}$ put the expression within the bracket = t .
- (8) For the type $(x+a)^{-1+\frac{1}{n}}(x+b)^{-1+\frac{1}{n}}$ or $\left(\frac{x+b}{x+a}\right)^{\frac{1}{n}-1} \frac{1}{(x+a)^2}$, put $\frac{x+b}{x+a} = t$.
- (9) For the type $\frac{1}{(x+a)^{n_1}(x+b)^{n_2}}$, put $(x+a) = t(x+b)$.
- (10) Integral of the form $F(x, \sqrt{ax^2 + bx + c})$ can be calculated by the following Euler's substitution:
- $\sqrt{ax^2 + bx + c} = t \pm x\sqrt{a}$ if $a > 0$.
 - $\sqrt{ax^2 + bx + c} = tx \pm \sqrt{c}$ if $c > 0$
 - $\sqrt{ax^2 + bx + c} = t(x-a)$ if $ax^2 + bx + c = a(x-a)(x-b)$.
- (11) Integral of the form $\frac{1}{ax^2 + bx + c}$ or $\frac{1}{\sqrt{ax^2 + bx + c}}$ or $\sqrt{ax^2 + bx + c}$ then represent $ax^2 + bx + c$ in the form of $A^2 + X^2$ or $A^2 - X^2$ or $X^2 - A^2$ where X is the linear expression in x and A is any constant and then integrate as per $\frac{1}{A^2 + X^2}$ or $\frac{1}{A^2 - X^2}$ or $\frac{1}{X^2 - A^2}$ or $\frac{1}{\sqrt{A^2 + X^2}}$ or $\frac{1}{\sqrt{A^2 - X^2}}$ or $\frac{1}{\sqrt{X^2 - A^2}}$ or $\sqrt{A^2 + X^2}$ or $\sqrt{A^2 - X^2}$ or $\sqrt{X^2 - A^2}$ and then integrate.
- (12) For $\int \frac{px+q}{ax^2+bx+c} dx$ or $\int \frac{px+q}{\sqrt{ax^2+bx+c}} dx$ or $\int (px+q)\sqrt{ax^2+bx+c} dx$ put $px+q = A \frac{d}{dx}(ax^2+bx+c) + B$ and $A = \frac{p}{2a}$, $B = q - bA$ then find the integral.
- (13) For $\int \frac{1}{(px+q)\sqrt{bx+c}} dx$ put $bx+c = t^2$ and solve.
- (14) For $\int \frac{1}{(px^2+qx+r)\sqrt{bx+c}} dx$ put $bx+c = t^2$ and solve.
- (15) For $\int \frac{1}{(px+q)\sqrt{ax^2+bx+c}} dx$ put $px+q = \frac{1}{t}$ and solve.
- (16) For $\int \frac{1}{(ax^2+b)\sqrt{cx^2+d}} dx$ put $x = \frac{1}{t}$ or $t^2 = \frac{cx^2+d}{ax^2+b}$ and solve.

- (17) For $\int \frac{\sqrt{ax+b}}{px+q} dx$ or $\int \frac{px+q}{\sqrt{ax+b}} dx$ or $\int (px+q)\sqrt{ax+b} dx$ put $ax+b=t^2$ and then find the integral.
- (18) For $\int \frac{1}{a^2 \cos^2 x + b^2 \sin^2 x} dx$ or $\int \frac{1}{a \cos^2 x + b \sin x \cos x + c \sin^2 x} dx$, divide the numerator and denominator by $\cos^2 x$ and then put $\tan x = t$ and solve.
- (19) $\int \frac{1}{a^2 \cos^2 x + b^2 \sin^2 x} dx = \frac{1}{ab} \tan^{-1} \left(\frac{b \sin x}{a \cos x} \right) + C.$
- (20) For $\int \frac{a \cos x + b \sin x + c}{d \cos x + e \sin x + f} dx,$
 Put $a \cos x + b \sin x + c = A \frac{d}{dx} (d \cos x + e \sin x + f) + B(d \cos x + e \sin x + f) + C$
- (21) $\int \frac{a \cos x + b \sin x}{c \cos x + d \sin x} dx = \frac{ac+bd}{c^2+d^2} + \frac{ad-bc}{c^2+d^2} \log |c \cos x + d \sin x| + C$
- (22) For $\int \frac{1}{a+b \sin x} dx$ or $\int \frac{1}{a+b \cos x} dx$ or $\int \frac{1}{a \cos x + b \sin x + c} dx$ put $\sin x = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$ and $\cos x = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$ and then put $\tan \frac{x}{2} = t$ and solve.
- (23) Integrals of the form $\int \frac{x^2+1}{x^4+kx^2+1} dx$ or $\int \frac{x^2-1}{x^4+kx^2+1} dx$ or $\int \frac{1}{x^4+kx^2+1} dx$
- (a) Divide Nr and Dr by x^2 we get
- (b) $\int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}+k} dx$ or $\int \frac{1-\frac{1}{x^2}}{x^2+\frac{1}{x^2}+k} dx$ or $\int \frac{\frac{1}{x^2}}{x^2+\frac{1}{x^2}+k} dx$
- (c) $\int \frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+2+k} dx$ or $\int \frac{1-\frac{1}{x^2}}{\left(x+\frac{1}{x}\right)^2-2+k} dx$ or $\frac{1}{2} \int \frac{\left(1+\frac{1}{x^2}\right) - \left(1-\frac{1}{x^2}\right)}{x^2+\frac{1}{x^2}+k} dx$
- (d) Put $x - \frac{1}{x} = u \Rightarrow \left(1 + \frac{1}{x^2}\right) dx = du$ and $x + \frac{1}{x} = v \Rightarrow \left(1 - \frac{1}{x^2}\right) dx = dv$
- (e) $\int \frac{du}{u^2 + (\sqrt{2+k})^2}$ or $\int \frac{dv}{v^2 + (\sqrt{-2+k})^2}$ or $\frac{1}{2} \int \frac{\left(1+\frac{1}{x^2}\right) - \left(1-\frac{1}{x^2}\right)}{x^2+\frac{1}{x^2}+k} dx$
- (f) Then Solve

- (24) Integrals of the form $\int \frac{x^2 + a^2}{x^4 + kx^2 + a^4} dx$ or $\int \frac{x^2 - a^2}{x^4 + kx^2 + a^4} dx$
- (a) Divide Nr and Dr by x^2 we get
- (b) $\int \frac{1 + \left(\frac{a}{x}\right)^2}{x^2 + \frac{a^4}{x^2} + k} dx$ or $\int \frac{1 - \left(\frac{a}{x}\right)^2}{x^2 + \frac{a^4}{x^2} + k} dx$
- (c) $\int \frac{1 + \left(\frac{a}{x}\right)^2}{\left(x - \frac{a^2}{x}\right)^2 + 2a^2 + k} dx$ or $\int \frac{1 - \left(\frac{a}{x}\right)^2}{\left(x + \frac{a^2}{x}\right)^2 - 2a^2 + k} dx$
- (d) Put $x - \frac{a^2}{x} = u \Rightarrow \left(1 + \frac{a^2}{x^2}\right) dx = du$ and $x + \frac{a^2}{x} = v \Rightarrow \left(1 - \frac{a^2}{x^2}\right) dx = dv$
- (e) $\int \frac{du}{u^2 + \left(\sqrt{2a^2 + k}\right)^2}$ or $\int \frac{dv}{v^2 + \left(\sqrt{-2a^2 + k}\right)^2}$
- (f) Then Solve.
- (25) Integral of the form $R[x, (ax+b)^{a/n}]$, where R is a rational function then put $ax+b = t^n$.
- (26) Integral of the form $R\left[x, \left(\frac{ax+b}{cx+d}\right)^{a/n}\right]$, where R is a rational function then put $\frac{ax+b}{cx+d} = t^n$.
- (27) Integral of the form $\int x^m (ax^n + b)^p dx$, where m, n, p are rational numbers:
- (a) If $p \in \mathbb{N}$, expand the integral with the help of binomial theorem and then integrate.
- (b) If $p \in \mathbb{Z}^-$ (negative integers), the integral reduces to the integral of rational function by means of the substitution $x = t^s$, where s is the LCM of m, n .
- (c) If $\frac{m+1}{n}$ is an integer, the integral can be rationalized by the substitution $ax^n + b = t^s$, where s is the denominator of the fraction P .
- (d) If $\frac{m+1}{n} + p$ is an integer, the integral can be rationalized by the substitution $ax^{-n} + b = t^s$, where s is the denominator of the fraction P .

(28) For $\int \frac{ax^2 + bx + c}{(dx + e)\sqrt{fx^2 + gx + h}} dx$ we have to rewrite :

$ax^2 + bx + c = A(dx + e)(2fx + g) + B(dx + e) + C$ and evaluate A, B and C by comparing the coefficient of like terms on both sides. Then integral can be written as:

$\int \frac{2fx + g}{\sqrt{fx^2 + gx + h}} dx + B \int \frac{1}{\sqrt{fx^2 + gx + h}} dx + C \int \frac{1}{(dx + e)\sqrt{fx^2 + gx + h}} dx$ and hence solve accordingly.

(29) Integral of the form $\int \frac{a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n}{\sqrt{ax^2 + bx + c}} dx$, assume that

$$Integral = (C_0x^{n-1} + C_1x^{n-2} + \dots + C_{n-1})\sqrt{ax^2 + bx + c} + C_n \int \frac{dx}{\sqrt{ax^2 + bx + c}} \text{ -----} \rightarrow (1),$$

where $C_0, C_1, C_2, \dots, C_n$ are constants, now differentiating (1) and by multiplying $\sqrt{ax^2 + bx + c}$ we get :

$$a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n = ((n-1)C_0x^{n-2} + (n-2)C_1x^{n-2} + \dots + C_{n-2})(ax^2 + bx + c) + (C_0x^{n-1} + C_1x^{n-2} + \dots + C_{n-1})(2ax + b)C_n$$

By comparing and equating the like terms find the value of $C_0, C_1, C_2, \dots, C_n$ and after substituting the values of $C_0, C_1, C_2, \dots, C_n$ in (1) we can integrate easily.
